Maximum Likelihood Estimation

How probability models produce common training losses

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Mathematical Foundations for AI · IIT Gandhinagar

Deriving losses from probability models

Three common supervised-learning losses

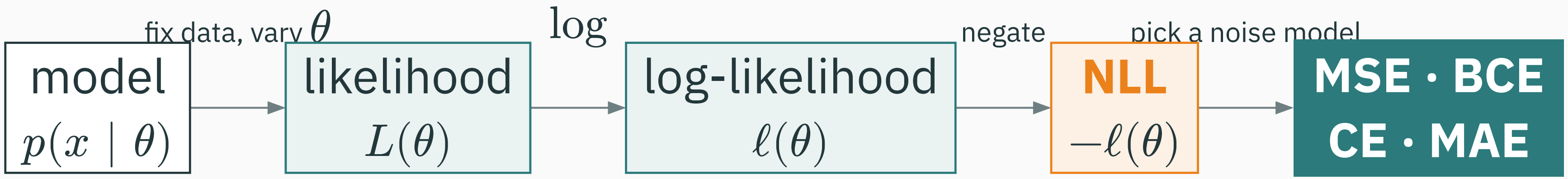
Many supervised-learning pipelines use one of these objectives:

```
loss = F.mse_loss(y_hat, y)              # regression
loss = F.binary_cross_entropy(p, y)      # yes/no classification
loss = F.cross_entropy(logits, y)        # multi-class classification
```

Squaring errors. Taking logs of probabilities. **Who decided these formulas?** Why squared — not absolute, not fourth power? Why is there a logarithm inside a classification loss?

> Each loss below is a negative log-likelihood for a particular observation model. Today we derive the construction and the Gaussian/MSE case; L24 returns to categorical cross-entropy.

From a probability model to an objective



Fix the observed data and vary the parameters. The following slides compute every step in this map.

> This lecture defines model fitting as optimizing an objective derived from an explicit statistical model.

Learning outcomes

By the end of this lecture you will be able to:

1. Read $p(x \mid \theta)$ **two ways**: as a distribution over x, and as a likelihood over θ.
2. Write the likelihood of an entire dataset using the **IID assumption**.
3. Explain **three independent reasons** the log goes in — and recap L2′s 0.03^{1000} underflow.
4. Derive $\hat{\theta}_{\text{MLE}}$ for the **Bernoulli** coin (⭐) and the **Gaussian** mean.
5. Derive the equivalence: **minimizing MSE = maximizing a fixed-variance Gaussian likelihood** (⭐).
6. Map noise models to the losses they generate: Gaussian→MSE, Bernoulli→BCE, Categorical→CE, Laplace→MAE.

The flip: from probability to likelihood

Pop quiz · one student, three courses

Three courses grade on curves. Their score distributions:

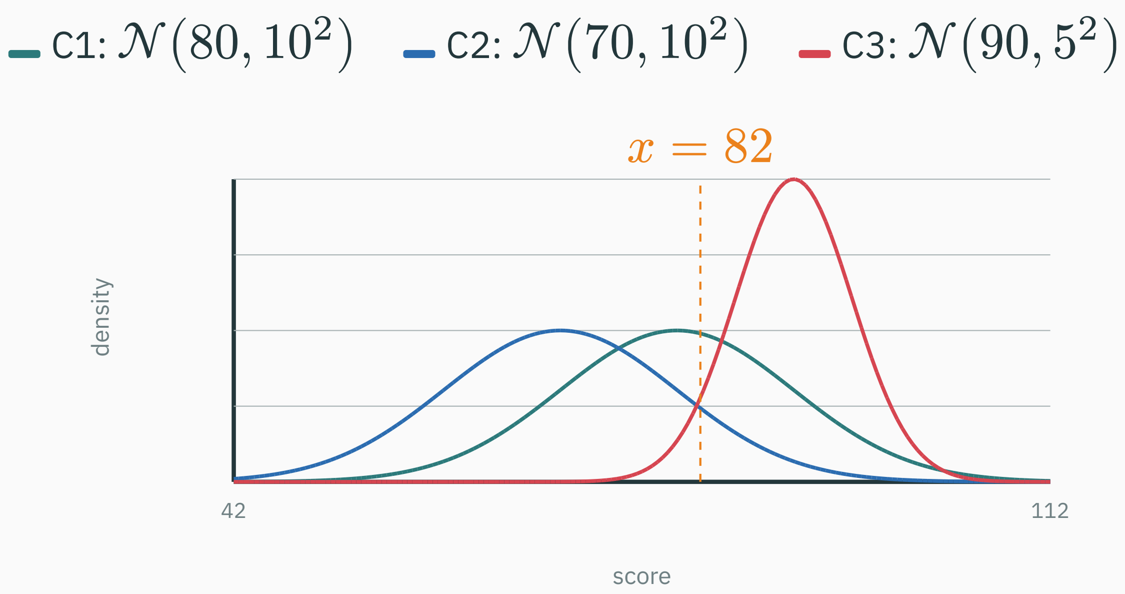
- C1: $\mathcal{N}(\mu = 80, \sigma = 10)$
- C2: $\mathcal{N}(\mu = 70, \sigma = 10)$
- C3: $\mathcal{N}(\mu = 90, \sigma = 5)$

I stop a random student in the corridor. They took exactly one of the three, and they scored **82**.

Which course would you guess? Commit to an answer — and notice what your brain just did to produce it.

Compare the three likelihoods

V VISUAL



Read each curve **at the observed score**: $p(82 \mid \mathrm{C1}) = 0.0391$, $p(82 \mid \mathrm{C2}) = 0.0194$, $p(82 \mid \mathrm{C3}) = 0.0222$.

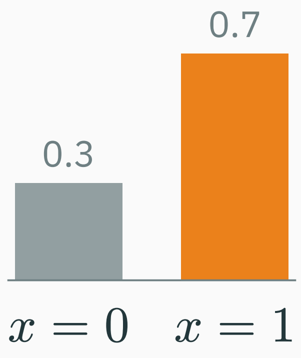
C1 wins: you picked the model under which the data is least surprising. That's maximum likelihood — you already do it by eye.

The flip, on the simplest model there is

V VISUAL

A coin with $P(\text{heads}) = \theta$. One formula, $p(x \mid \theta)$ — **two completely different readings**:

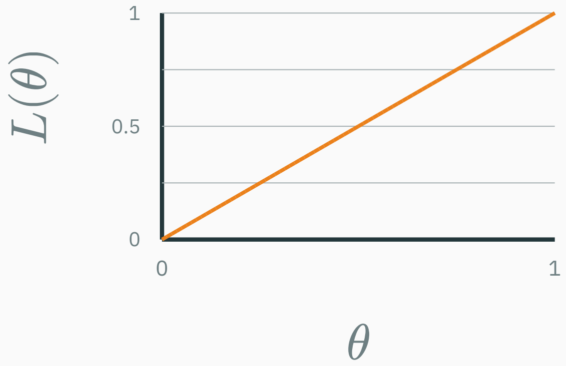
Fix $\theta = 0.7$, vary x



a **distribution** over outcomes

bars sum to 1 — guaranteed

Fix $x = 1$ (one head), vary θ



the **likelihood** of that outcome

area under this line: $1/2$ — nothing forces it to be 1

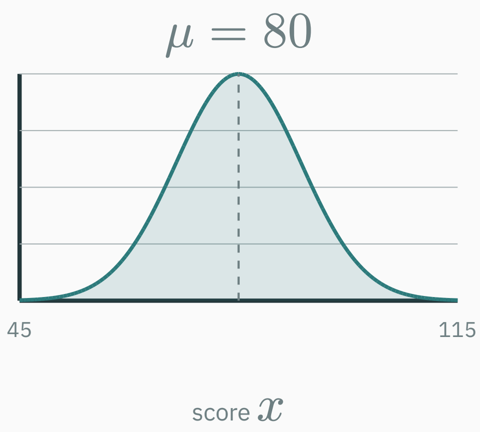
Two bars became a line: **different axis, different object** — same formula.

The flip, on the corridor student

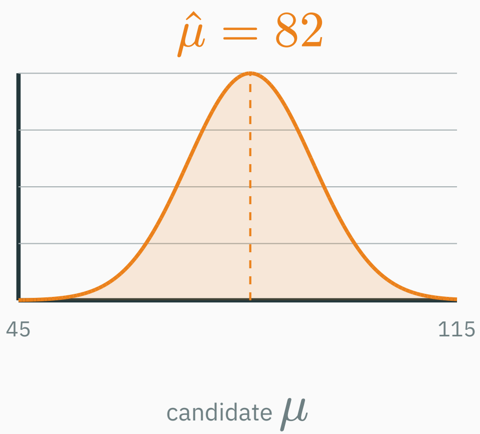
V VISUAL

Now suppose we know the course has $\sigma = 10$ but **nobody told us μ**. The student says 82.

DISTRIBUTION: fix $\mu = 80$, vary x



LIKELIHOOD: fix $x = 82$, vary μ



The right curve peaks at $\mu = 82$: the single best explanation centers the bell **on the data**. You just estimated a parameter.

Naming the move

$L(\theta) := p(\mathcal{D} \mid \theta)$**, read as a function of θ with the data $\mathcal{D}$ frozen.**

$$\hat{\theta}_{\text{MLE}} = \arg\max_\theta L(\theta).$$

- $p(x \mid \theta)$ as a function of x: a **distribution** — normalized, sums/integrates to 1
- the same formula as a function of θ: the **likelihood** — a plain score, one number per candidate model

$L(\theta)$ is **not** a probability distribution over θ: nothing makes its area 1 (the one-head line had area $1/2$). "Which θ is most probable" is a different, stronger question — that's Bayes and L15. Today we only ask which θ **explains the data best**.

The likelihood of a dataset

From one observation to a dataset

One data point gives one score: $L(\theta) = p(x_1 \mid \theta)$. A dataset $\mathcal{D} = \{x_1, \ldots, x_n\}$ needs the **joint** probability $p(x_1, \ldots, x_n \mid \theta)$ — and joints are hard. We buy simplicity with an assumption. **IID** is two separate claims:

- **Independent** — given θ, no observation tells you about another → the joint **factors into a product**
- **Identically distributed** — every observation uses the **same** $p(\cdot \mid \theta)$, one shared θ

$$L(\theta) = \prod_{i=1}^{n} p(x_i \mid \theta)$$

Each claim can fail alone: tomorrow's weather depends on today's (→ L25′s Markov chains); two different-noise sensors share no rule. IID is a modeling **choice** — say it out loud.

IID at work · two coins, two tiny datasets

Candidate coins: fair ($\theta = 0.5$) vs head-heavy ($\theta = 0.8$). Score both on data:

Observed	$L(0.5)$	$L(0.8)$	Better explanation
H, T	$0.5 \times 0.5 = 0.25$	$0.8 \times 0.2 = 0.16$	the fair coin
H, H	$0.5^2 = 0.25$	$0.8^2 = 0.64$	the head-heavy coin

Same two candidates, different frozen data → different winner. The likelihood **ranks models by the data they actually saw**.

Independence licenses the multiplication; identical distribution licenses reusing one θ in every factor.

Pop quiz, round 2 · now with a dataset

Back in the corridor. **Two** students this time — scores 82 and 72 — and both took the **same** (unknown) course. Multiply each course's two density readings:

Course	$p(82 \mid \cdot)$	$p(72 \mid \cdot)$	$L =$ **product**
C1: $\mathcal{N}(80, 10^2)$	0.0391	0.029	1.1×10^{-3}
C2: $\mathcal{N}(70, 10^2)$	0.0194	0.0391	7.6×10^{-4}
C3: $\mathcal{N}(90, 5^2)$	0.0222	0.0001	2.7×10^{-6}

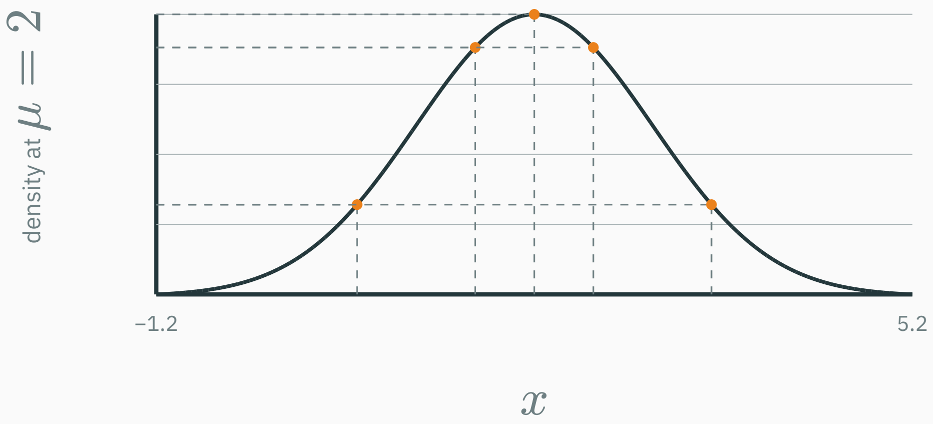
C1 wins again — but look at C3: **one** far-off point (72, at $z = -3.6$) collapsed its entire product. A product is a chain of vetoes: every point must be plausible.

A dataset's likelihood multiplies one reading per point — computed with the IID recipe you just learned.

Five readings, one score

V VISUAL

Our running dataset: $\mathcal{D} = \{0.5, 1.5, 2.0, 2.5, 3.5\}$, modeled as $\mathcal{N}(\mu, 1)$ with σ known. Try $\mu = 2$: read the density at each point, multiply.



$$L(2) = 0.13 \times 0.352 \times 0.399 \times 0.352 \times 0.13 \approx 8.3 \times 10^{-4}$$

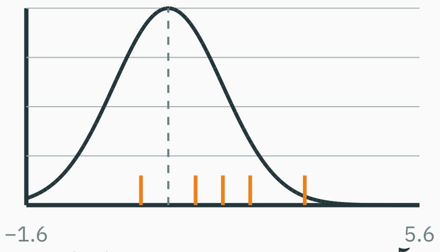
One number for the candidate $\mu = 2$. Now the key mental animation: **slide the bell and watch the score move**.

Slide the Gaussian over the data

V VISUAL

too far left

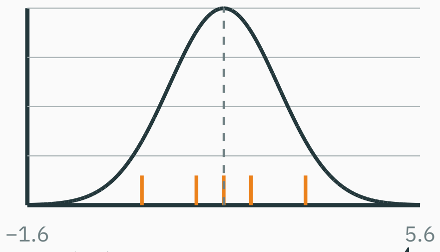
$\mu = 1$



$L(1) \approx 6.8 \times 10^{-5}$

just right

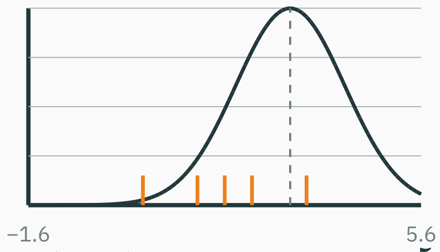
$\mu = 2$



$L(2) \approx 8.3 \times 10^{-4}$

too far right

$\mu = 3.2$



$L(3.2) \approx 2.3 \times 10^{-5}$

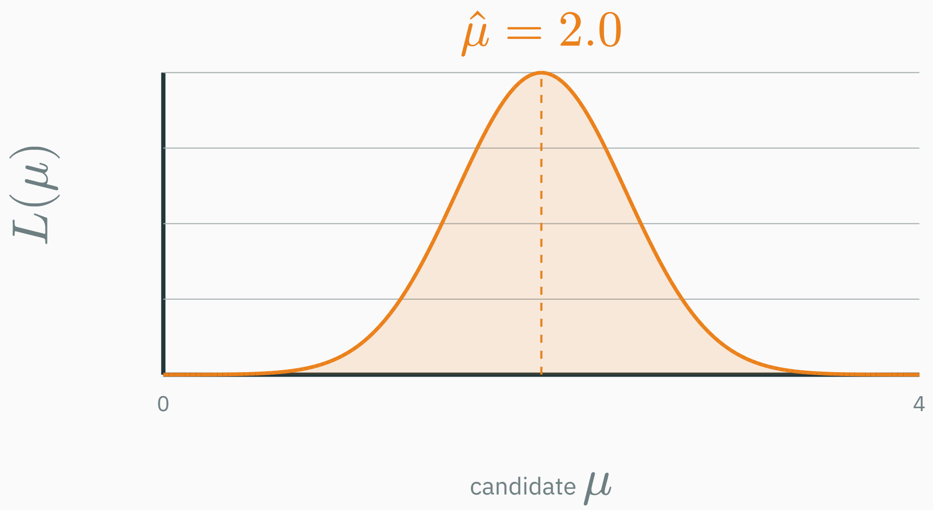
Same five orange data points every time; only the model slides. Centered wins by a factor of **12×** over $\mu = 1$.

Fitting a model = sliding its parameters until the data's likelihood is maximal.

The whole slide, as one curve

V VISUAL

Evaluate $L(\mu)$ at **every** candidate μ — the likelihood function of this dataset:



- Each point of this curve is one “slide position” from the previous slide — a **product of five densities**
- The peak sits at $\hat{\mu} = 2.0$ — which happens to be the sample mean of $\mathcal{D}$. Coincidence? We’ll **prove** it shortly.

Checkpoint: what kind of object is $L(\mu)$?

Q QUESTION

The curve $L(\mu)$ we just drew for $\mathcal{D} = \{0.5, 1.5, 2.0, 2.5, 3.5\}$ — which statement is true?

A. It is a probability density over μ, so its area must be 1

B. $L(2.0)$ is the probability that $\mu = 2.0$ is the true parameter

C. $L(\mu)$ is the joint density of the five **observed** points, re-read as a function of μ

D. Its peak location depends on the order in which the five points were multiplied

Answer: what kind of object is $L(\mu)$?

A ANSWER

Correct: C — the joint density of the observed data, as a function of μ

Why: The data is frozen; μ varies. Nothing normalizes the curve over μ (A), and no probability is ever assigned **to** μ — that requires a prior and Bayes' rule, which is L15′s business (B). Multiplication is commutative, so order is irrelevant (D).

Log-likelihood: products become sums

Reason 1 · products of probabilities underflow

You proved this in L2 — the recap: a language model scores a 1000-character text, each character with probability ≈ 0.03:

$$P(\text{text}) = \prod_{i=1}^{1000} p_i \approx 0.03^{1000} \approx 10^{-1523} \underset{\text{even float64}}{\Rightarrow} \mathbf{0.0} \quad \text{argmax of all-zeros = garbage}$$

Our five-point likelihood already sat at 10^{-4}. In log-space, tiny becomes ordinary:

$$\log P(\text{text}) = \sum_{i=1}^{1000} \log p_i \approx 1000 \times (-3.51) = -3507 \quad \checkmark \text{ a boring, safe float}$$

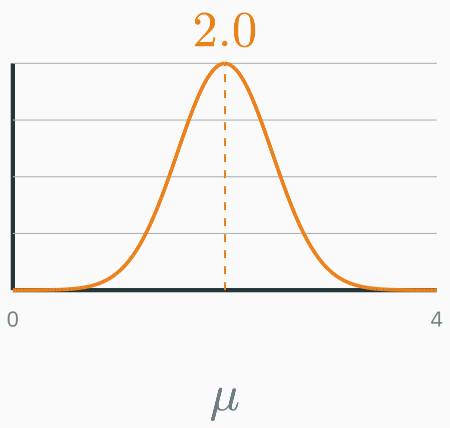
Log space prevents underflow in likelihood products. It also preserves the maximizer and turns independent per-example factors into an additive objective.

Reason 2 · the log moves the peak nowhere

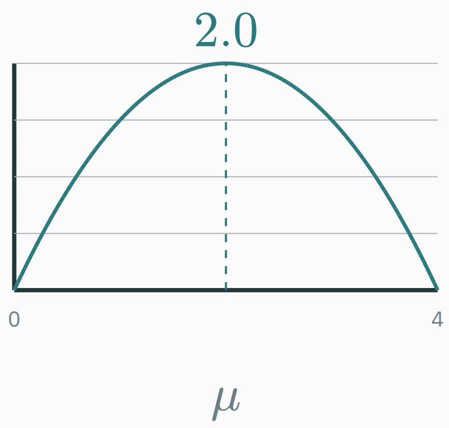
V VISUAL

$\log$ is **strictly increasing**: $a > b > 0 \Rightarrow \log a > \log b$. So it preserves every comparison of likelihoods — including where the maximum sits:

LIKELIHOOD $L(\mu)$



LOG-LIKELIHOOD $\ell(\mu)$



$$\arg\max_{\theta} L(\theta) = \arg\max_{\theta} \log L(\theta)$$

— same peak, tamer numbers

Reason 3 · sums are what calculus (and autodiff) want

$$\ell(\theta) := \log L(\theta) = \log \prod_{i=1}^{n} p(x_i \mid \theta) = \sum_{i=1}^{n} \log p(x_i \mid \theta)$$

To maximize, we differentiate (L7–L8) and set to zero. Compare the two forms:

- $\frac{\mathrm{d}}{\mathrm{d}\theta} \prod_i p_i$: the product rule over n factors — n terms, each a product of $n-1$ densities
- $\frac{\mathrm{d}}{\mathrm{d}\theta} \sum_i \log p_i = \sum_i \left(\frac{\mathrm{d}}{\mathrm{d}\theta}\right) \log p_i$: differentiate each term **alone**, add

For likelihood-based training, logs prevent underflow, preserve the argmax, and produce a sum of per-example terms. Other objectives also exist and need not be log-likelihoods.

Negative log-likelihood as a minimization objective

Optimizers by convention **minimize** (Module 4 is built that way). So negate:

$$\mathrm{NLL}(\theta) = -\ell(\theta) = -\sum_{i=1}^{n} \log p(x_i \mid \theta) \qquad \text{and often the mean} \quad \frac{1}{n}\sum_{i} -\log p(x_i \mid \theta)$$

- maximize likelihood $\Leftrightarrow$ maximize log-likelihood $\Leftrightarrow$ **minimize NLL**
- each data point contributes $-\log p(x_i \mid \theta)$: **its personal penalty** — large when the model found it implausible

Training loss = per-example negative log-likelihood, averaged. The rest of the lecture instantiates this construction for specific observation models.

The coin: your first MLE, derived ⭐

The model, and a clever way to write it

Flip a coin 10 times: $H, H, T, H, H, H, T, H, H, T$ — **7 heads, 3 tails**. Model: each flip is Bernoulli, $P(H) = \theta$, IID. Code heads as $x = 1$, tails as $x = 0$:

$$p(x \mid \theta) = \theta^x (1-\theta)^{1-x} \qquad \text{one formula, both cases: } x = 1 \to \theta;\ x = 0 \to 1-\theta$$

The dataset's likelihood — products of θ's and $(1-\theta)$'s, one per flip:

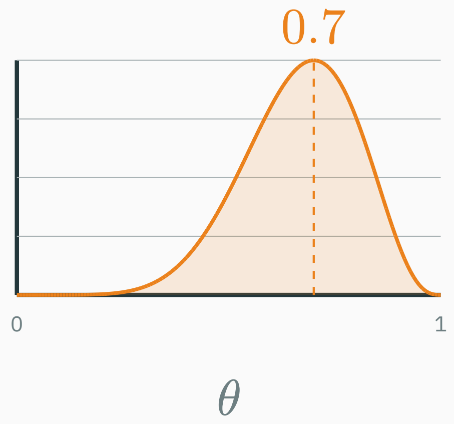
$$L(\theta) = \prod_{i=1}^{10} \theta^{x_i} (1-\theta)^{1-x_i} = \theta^7 (1-\theta)^3$$

Only the **counts** survive: 7 heads, 3 tails. The order of flips never mattered — IID told us so before we computed anything.

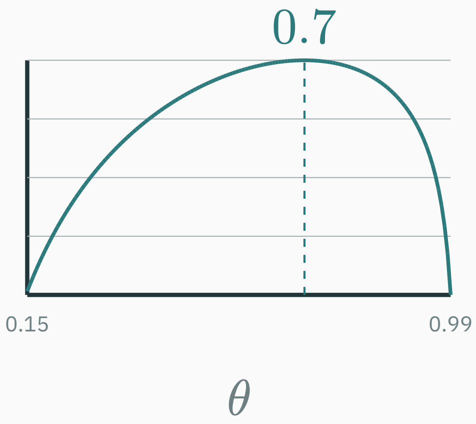
The two curves, computed

V VISUAL

$L(\theta) = \theta^7(1-\theta)^3$



$\ell(\theta) = 7\log\theta + 3\log(1-\theta)$



Both peak at $\theta = 0.7$. Spot values: $L(0.5) = 0.5^{10} \approx 9.8 \times 10^{-4}$, $L(0.7) \approx 2.2 \times 10^{-3}$ — the data is 2.3× more plausible under $\theta = 0.7$ than under a fair coin.

⭐ Derivation · set the derivative to zero

D DERIVATION

Maximize $\ell(\theta) = 7\log\theta + 3\log(1-\theta)$ the L7 way — stationary point:

$$\frac{\mathrm{d}\ell}{\mathrm{d}\theta} = \frac{7}{\theta} - \frac{3}{1-\theta} = 0$$

$$7(1-\theta) = 3\theta \quad \Rightarrow \quad 7 = 10\theta \quad \Rightarrow \quad \hat{\theta} = \frac{7}{10} = 0.7$$

Is it a maximum? Check curvature (L7′s second-derivative test):

$$\frac{\mathrm{d}^2\ell}{\mathrm{d}\theta^2} = -\frac{7}{\theta^2} - \frac{3}{(1-\theta)^2} < 0 \quad \text{everywhere on } (0,1)$$

Strictly concave → the single stationary point is the **global** maximum. No luck involved.

⭐ Derivation · now for any dataset

D DERIVATION

General case: n flips, n_H heads, $n_T = n - n_H$ tails. Same three moves:

$$\ell(\theta) = \sum_{i=1}^{n} [x_i \log \theta + (1 - x_i) \log(1 - \theta)] = n_H \log \theta + n_T \log(1 - \theta)$$

$$\frac{d\ell}{d\theta} = \frac{n_H}{\theta} - \frac{n_T}{1 - \theta} = 0 \quad \Rightarrow \quad n_H(1 - \theta) = n_T \theta$$

$$n_H = (n_H + n_T)\theta$$

$\hat{\theta}_{\text{MLE}} = \frac{n_H}{n}$ **— the fraction of heads.**

The answer your intuition would have guessed — but now it is a **theorem**, from a principle that will still work when intuition has nothing to say (a billion-parameter model has no “fraction of heads”).

Two checks on the Bernoulli estimate

The machine agrees — brute-force the curve:

```
heads, n = 7, 10
theta  = np.linspace(0.01, 0.99, 981)
loglik = heads * np.log(theta) + (n - heads) * np.log(1 - theta)
round(theta[np.argmax(loglik)], 3)          # 0.7  — the head fraction
```

Now flip a coin **3 times**, get **3 heads**: $\hat{\theta}_{\text{MLE}} = 3/3 = 1.0$ — this coin **cannot ever** land tails. Three flips convinced MLE of a certainty. Your intuition says “probably just a normal coin”. Your intuition is using a **prior** — and giving MLE one is exactly L15.

Interactive · climb the coin's likelihood yourself

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/mle-map-coin

The **mle-map-coin** article flips coins live: the likelihood curve rebuilds after every flip, and you drag θ to feel the score change under your cursor. Recreate today's 7-of-10, then try 3-of-3 and watch the peak slam into the boundary at $\theta = 1$. (Ignore the prior slider for now — after L15 it will be the whole point.)

Two-minute experiment before next lecture: keep the head **fraction** at 70% but grow n: 7/10 → 70/100 → 700/1000. Watch what happens to the curve's **width** — we name that phenomenon at the end of today.

MLE for the Gaussian

The Gaussian log-likelihood, simplified once and for all

Model $\mathcal{D} = \{x_1, ..., x_n\}$ as IID $\mathcal{N}(\mu, \sigma^2)$ (L12′s density, L13′s star):

$$\ell(\mu, \sigma) = \sum_{i=1}^{n} \log\left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)\right]$$

$\log$ of (product of exp) — everything unfolds:

$$\ell(\mu, \sigma) = -\frac{n}{2}\log(2\pi) - n\log\sigma - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2$$

All the μ-dependence sits in one term: $-\sum_i (x_i - \mu)^2$. Maximizing ℓ over μ is minimizing a sum of squares. Remember this shape — it returns in ten minutes as MSE.

The mean, earned

Differentiate w.r.t. μ (the constants and $-n\log\sigma$ vanish):

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0 \quad \Rightarrow \quad \sum_i x_i = n\mu$$

$$\hat{\mu}_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

On our running dataset: $\hat{\mu} = (0.5 + 1.5 + 2.0 + 2.5 + 3.5)/5 = 2.0$ — **exactly where the likelihood curve peaked** in Section 3. The picture and the calculus agree.

Note what $\hat{\mu}$ does **not** depend on: σ. However noisy the ruler, the best center is the average. (With σ unknown, the μ-equation is unchanged.)

The variance and its finite-sample bias

Same recipe for σ: differentiate $\ell = \ldots - n\log\sigma - \frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2$, set to zero, solve:

$$\frac{\partial \ell}{\partial \sigma} = -\frac{n}{\sigma} + \frac{1}{\sigma^3}\sum_i (x_i - \hat{\mu})^2 = 0 \quad \Rightarrow \quad \hat{\sigma}^2_{\mathrm{MLE}} = \frac{1}{n}\sum_{i=1}^{n} (x_i - \bar{x})^2$$

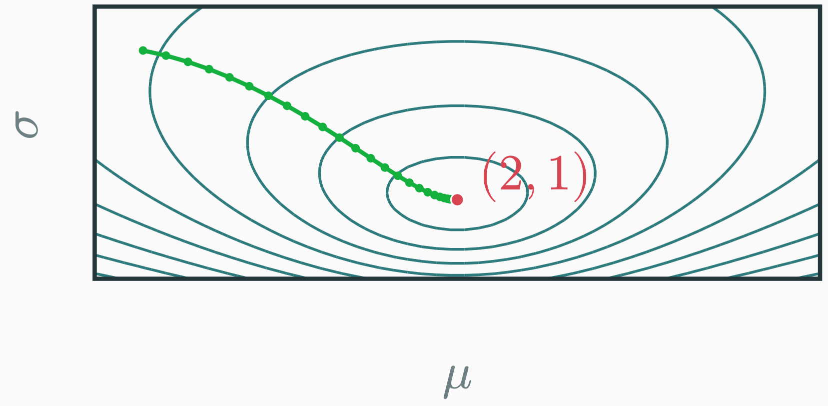
On $\mathcal{D}$: deviations $(-1.5, -0.5, 0, 0.5, 1.5)$, squares sum to 5.0, so $\hat{\sigma}^2 = 5.0/5 = 1.0$.

This MLE is biased: across repeated datasets, $\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \cdot \sigma^2$. NumPy's `np.var(x, ddof=1)` applies Bessel's $\frac{n}{n-1}$ correction to obtain an unbiased estimator under Gaussian sampling.

Estimation is optimization · the NLL landscape

V VISUAL

Free both parameters: the **dataset NLL** over the (μ, σ) plane, computed from $\mathcal{D}$:



- The bowl bottoms out at $(\hat{\mu}, \hat{\sigma}) = (2.0, 1.0)$ — both closed forms at once
- The green path: **gradient descent** walking downhill on this exact surface

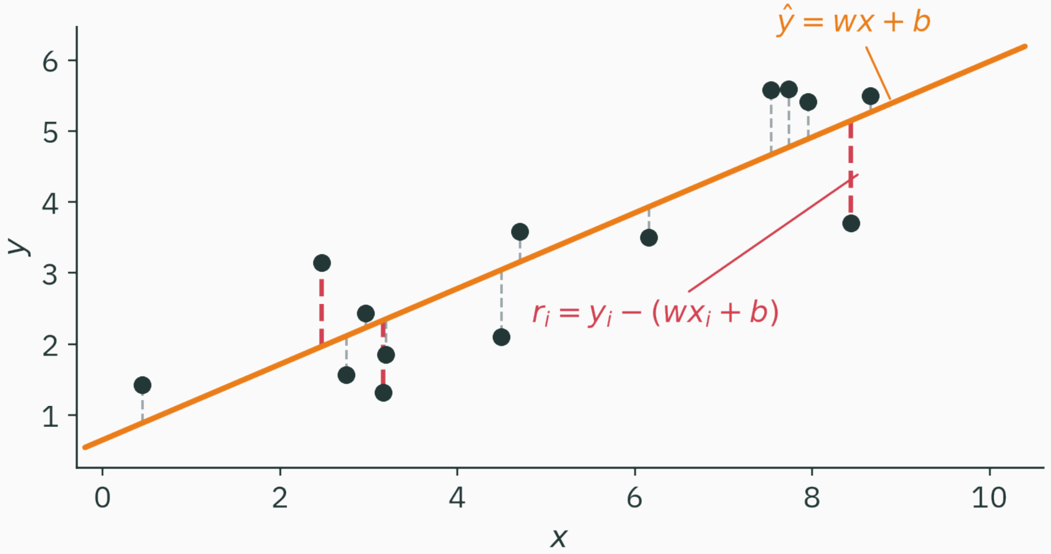
Fitting = minimizing NLL. Module 4 (L16–L21) is the art of walking downhill on such surfaces.

Linear regression with Gaussian noise → least squares

A model that finally has an input

V VISUAL

Everything so far fit a **constant** distribution. Real problems predict y **from** x — rent from area, temperature from hour. The simplest such model is a line:



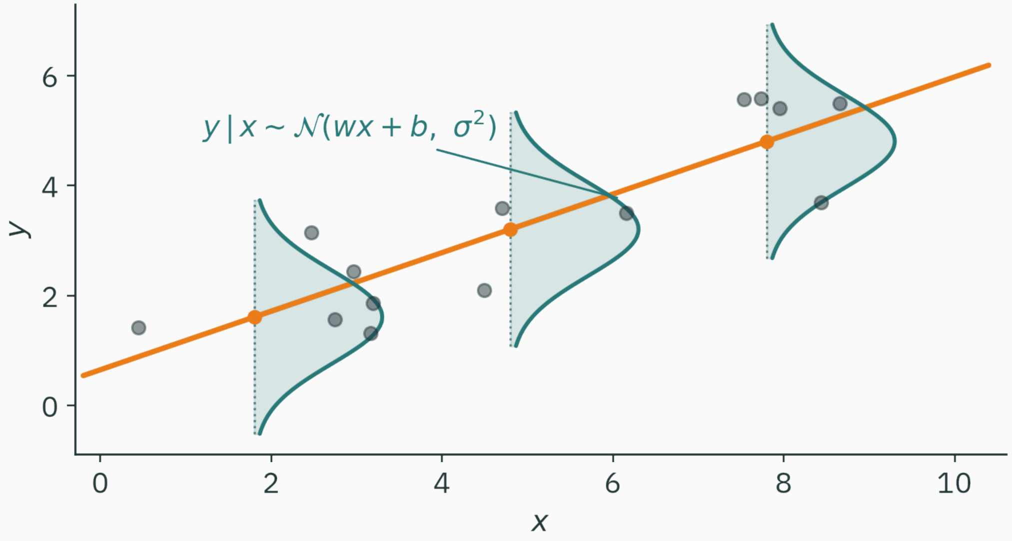
$\hat{y} = wx + b$; the **residual** $r_i = y_i - (wx_i + b)$ is what the line cannot explain. Which line is **best**? We need a likelihood — so we need a **distribution over y**.

The noise model: a bell standing on the line

V VISUAL

Assume the world draws y as **the line plus Gaussian noise**:

$$y_i = wx_i + b + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2) \text{ IID} \quad \Rightarrow \quad y_i \mid x_i \sim \mathcal{N}(wx_i + b, \sigma^2)$$



A vertical Gaussian stands **on the line** at every x: near the line, high density; stragglers, expensive. That is the whole assumption — now we turn the crank.

The likelihood of a **labeled** dataset

New situation, same machinery. The data comes in pairs (x_i, y_i), and our model only says how y arises **given** x — it never models the inputs:

$$L(\theta) = \prod_{i=1}^{n} p(y_i \mid x_i, \theta) \qquad \theta = (w, b);\ \text{the } x_i \text{ enter as fixed givens}$$

- Same flip: data (x_i, y_i) frozen, parameters (w, b) varying
- Same IID product — independence **of the noise** ε_i, one shared line
- No $p(x)$ anywhere: whoever chose the inputs is not our problem

This conditional form is the shape of **every** supervised objective — including the trillion-token ones: L26′s language model maximizes exactly such a product, one conditional per next character.

⭐ Derivation · one point's log-likelihood

D DERIVATION

The density of observing label y_i at input x_i, given parameters $\theta = (w, b)$:

$$p(y_i \mid x_i, \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - (wx_i + b))^2}{2\sigma^2} \right)$$

Take the log — the same unfold as the Gaussian section:

$$\log p(y_i \mid x_i, \theta) = -\frac{(y_i - (wx_i + b))^2}{2\sigma^2} - \log\sigma - \frac{1}{2}\log(2\pi)$$

$$-\log p(y_i \mid x_i, \theta) = \frac{\overbrace{(y_i - \hat{y}_i)^2}^{}{}_{\text{squared residual!}}}{2\sigma^2} + \underbrace{\log\sigma + \frac{1}{2}\log(2\pi)}_{\text{no } \theta \text{ inside — constant}}$$

One data point's NLL **is** its squared residual, rescaled and shifted by constants.

⭐ Derivation · sum, strip constants, done

D DERIVATION

Sum over the IID dataset and maximize over $\theta = (w, b)$, with σ fixed:

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \ell(\theta) = \arg\min_{\theta} \sum_{i=1}^{n} \left[\frac{(y_i - \hat{y}_i)^2}{2\sigma^2} + \log \sigma + \frac{1}{2} \log(2\pi) \right]$$

Adding a constant never moves an argmin — drop $n \log \sigma + \frac{n}{2} \log(2\pi)$. Scaling by a positive constant never moves an argmin — drop $1/(2\sigma^2)$:

$\hat{\theta}_{\mathrm{MLE}} = \arg\min_{w,b} \sum_{i=1}^{n} (y_i - wx_i - b)^2$. With fixed-variance Gaussian observation noise, maximum likelihood and least squares have the same minimizers.

A likelihood interpretation of MSE

If MSE is interpreted as a negative log-likelihood, the corresponding observation model makes three assumptions:

1. noise is **Gaussian** — bell-shaped errors around the prediction
2. noise has **the same σ everywhere** — no input is harder than another
3. observations are **independent** given the inputs

The quadratic penalty gives outliers high leverage: a point 5σ from the line contributes $(5\sigma)^2/2\sigma^2 = 12.5$ nats. A heavier-tailed observation model, such as Laplace, yields a linear absolute-error penalty.

The machine agrees · one grid, two objectives

```
x  = np.array([0., 1., 2., 3., 4.]); y = np.array([1.1, 1.9, 2.8, 4.3, 4.9])
ws = np.linspace(0.2, 1.8, 801)                 # candidate slopes (b = 1 fixed)
mse = [np.mean((y - w*x - 1.0)**2) for w in ws]
ll  = [np.sum(-0.5*np.log(2*np.pi) - 0.5*(y - w*x - 1.0)**2) for w in ws]
print(ws[np.argmin(mse)], ws[np.argmax(ll)])    # 1.0 1.0  — the same slope
```

With fixed σ, these two objectives have the same minimizer for any dataset because they differ only by a positive scale and an additive constant.

Least squares can be introduced geometrically or as Gaussian maximum likelihood. The likelihood view makes the observation-noise assumptions explicit.

Common losses as negative log-likelihoods

Match an observation model to its NLL

Choose a noise/observation model; the NLL **is** your loss:

You observe	Model	NLL per example	Its famous name
a real y	Gaussian	$(y-\hat{y})^2/2\sigma^2 + c$	**MSE** ✓ today
yes / no	Bernoulli	$-[y\log p + (1-y)\log(1-p)]$	**BCE** ✓ today
1 of K classes	Categorical	$-\log p_{\text{true class}}$	**cross-entropy** → L24
y with outliers	Laplace	$\lvert y-\hat{y}\rvert/b + c$	**MAE** bonus

These common losses share one construction: choose an observation model and minimize its negative log-likelihood. This table is not a claim that every possible objective is an NLL.

The BCE row costs one line

For a yes/no label $y \in \{0, 1\}$ with predicted head-probability p, the Bernoulli pmf in exponent form (from the coin section!) gives the per-example NLL:

$$-\log\left[p^y(1-p)^{1-y}\right] = -\left[y\log p + (1-y)\log(1-p)\right]$$

This is `binary_cross_entropy`, derived directly from the Bernoulli log-likelihood.

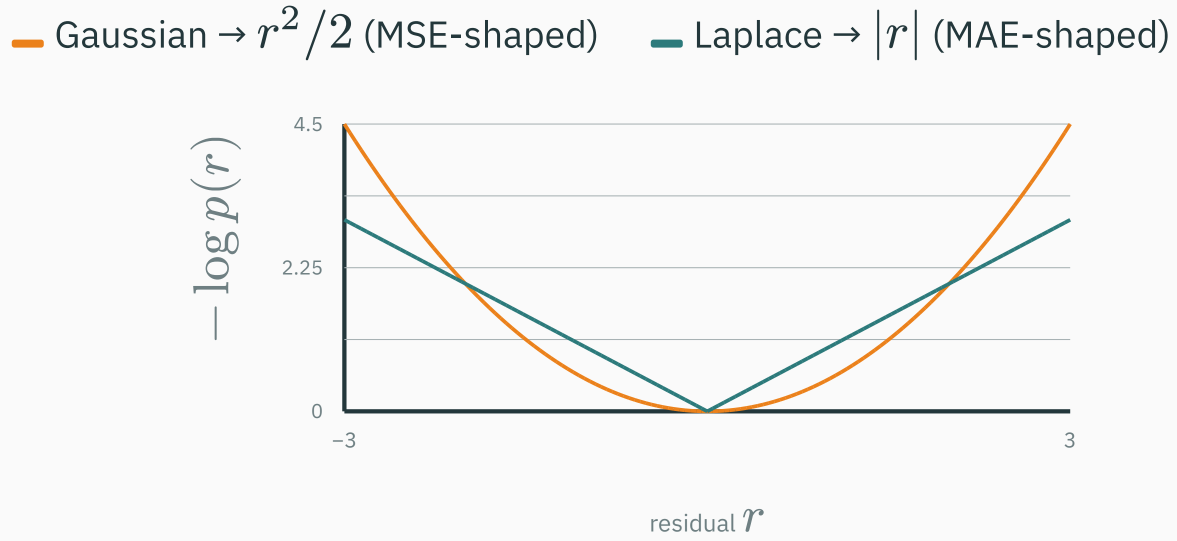
- $y = 1$, model said $p = 0.9$: penalty $-\log 0.9 \approx 0.105$ — cheap
- $y = 1$, model said $p = 0.01$: penalty $-\log 0.01 \approx 4.6$ — confident wrongness is **expensive**, by the shape of $-\log$

The categorical / cross-entropy row is the same move over K classes — L24 does it properly and connects it to information theory (why “entropy” is in the name), and L2′s stable log-softmax is exactly how it’s computed.

Two noise models, two loss shapes

V VISUAL

The **true** per-residual loss curves, computed from the densities themselves ($-\log p$, shifted to 0 at $r = 0$):



For the Gaussian NLL, doubling a residual quadruples its cost, so outliers have high leverage. The Laplace NLL grows linearly with $|r|$, reducing that leverage. The loss shape encodes a particular residual model.

Match each common loss to an observation model

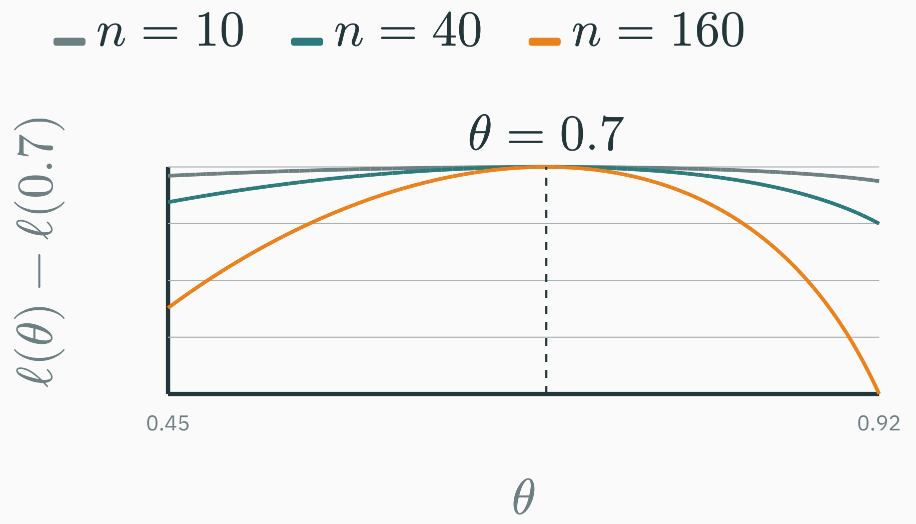
Loss	**Where it comes from**	**Derived**
`mse_loss`	Gaussian noise on the prediction — the ⭐ derivation above	**today**
`binary_cross_entropy`	Bernoulli NLL — the coin's likelihood, written per example	**today**
`cross_entropy`	Categorical NLL + information theory: **why it's called entropy**, what it has to do with compression	**→ L24**

L2 told you `CrossEntropyLoss` fuses log-softmax for stability and takes raw logits. Today you learned why it's a **loss** at all. L24 explains why it's called **entropy**. Three lectures, one function — that's how central it is.

Properties, in brief · more data sharpens the peak

V VISUAL

Hold the head fraction at 70%, grow n — the peak-aligned log-likelihood sharpens:



- **Consistency** (stated, under correct specification, identifiability, and regularity conditions): $\hat{\theta}_{\mathrm{MLE}}$ converges to the true parameter as $n \to \infty$
- Small samples expose limitations: $\hat{\sigma}^2$ is biased and three identical flips produce a boundary estimate. L15 shows how a prior changes the estimator and represents uncertainty.

Checkpoint: choose the loss like a modeler

Q QUESTION

Your sensor's errors are usually small but show occasional huge spikes — well modeled by Laplace noise. Maximum likelihood hands you which training loss?

A. MSE — squared error is always optimal

B. MAE — the mean **absolute** error

C. BCE — it has a log, logs handle spikes

D. Cross-entropy over discretized bins

Answer: choose the loss like a modeler

A ANSWER

Correct: B — MAE — mean absolute error

Why: Laplace density $\propto e^{-|r|/b}$, so per-point NLL $= |r|/b + c$: the absolute residual. The quadratic MSE would let each spike bully the fit (r^2 explodes); the linear penalty stays calm. Same construction as MSE — different noise belief. That is the table doing real work.

Practice problems

Try on paper; the T7 notebook (after L15) checks all of them in NumPy.

1. 12 flips, 9 heads: write $\ell(\theta)$, maximize it, and compute $\ell(0.75) - \ell(0.5)$.
2. For $\mathcal{D} = \{1.0, 3.0\}$ under $\mathcal{N}(\mu, \sigma^2)$: sketch $L(\mu)$ at $\sigma = 1$; find $\hat{\mu}$, then $\hat{\sigma}^2$. (Answers: 2.0 and 1.0.)
3. Exponential waiting times $p(x \mid \lambda) = \lambda e^{-\lambda x}$ (L12's zoo): derive $\hat{\lambda}_{\text{MLE}} = 1/\bar{x}$.
4. Repeat the derivation with **Laplace** noise and show that the objective becomes $\sum_i |y_i - wx_i - b|$.
5. For $n = 10^6$, $p_i \approx 0.1$: estimate $\prod_i p_i$ and compare with float64's smallest positive subnormal ($\approx 5 \times 10^{-324}$, L2). Why must training code never form this product?
6. True or false, one sentence: "the MLE of θ is the most probable value of θ given the data." Explain the distinction introduced in L15.

Lecture 14 — summary

- **The flip**: freeze the data, vary θ — the likelihood is a score, not a distribution over θ.
- **Datasets**: IID → a product of per-point densities; slide the model, climb the score.
- **Logs**: same argmax, no underflow, and additive per-point derivatives. NLL gives many standard losses.
- **Coin ⭐**: $\hat{\theta} = n_H/n$ from $\ell'(\theta) = 0$; 3-for-3 heads exposes MLE's blind spot (→ L15).
- **Gaussian**: $\hat{\mu} = \bar{x}$; $\hat{\sigma}^2 = \frac{1}{n}\sum(x_i - \bar{x})^2$ — the variance MLE is biased at finite n.
- **Linear regression ⭐**: fixed-variance Gaussian noise → maximum likelihood $\equiv$ minimum squared error.
- **The table**: Gaussian→MSE, Bernoulli→BCE, Categorical→CE (L24), Laplace→MAE.

Lecture 14 — summary

Read before L15 — MML §8.1–8.3 (skim 8.1–8.2, focus 8.3) · MacKay Ch. 2. **T7** (after L15) drills MLE + MAP together.

⭐⭐⭐ Where MLE points as $n \to \infty$

★ OPTIONAL

Divide the log-likelihood by n: an **average** that (by large numbers) approaches an expectation over the true data source $p^\star$:

$$\frac{1}{n}\ell(\theta) = \frac{1}{n}\sum_{i=1}^{n} \log p(x_i \mid \theta) \quad \xrightarrow[n\to\infty]{} \quad \mathbb{E}_{x\sim p^\star}[\log p(x \mid \theta)]$$

Add and subtract the entropy of $p^\star$ (L22 defines it properly):

$$\mathbb{E}_{p^\star}[\log p(x \mid \theta)] = \underbrace{\mathbb{E}_{p^\star}[\log p^\star(x)]}_{\text{fixed: no } \theta} - \underbrace{\mathrm{KL}(p^\star \,\|\, p_\theta)}_{>= 0\text{, gap to reality}}$$

Maximizing likelihood is equivalent to minimizing $\mathrm{KL}(p^\star \parallel p_\theta)$ up to a constant independent of θ. L24 derives this identity from cross-entropy.

Many standard supervised losses are negative log-likelihoods for explicit observation models.

Next: **MAP & Conjugate Priors** — what if you have beliefs *before* seeing data?